Enrollment No. 01601172024

MID-TERM EXAMINATION
(Course Name : MTech CSE AI) (Semester 2)
(February, 2023) OFF LINE mode

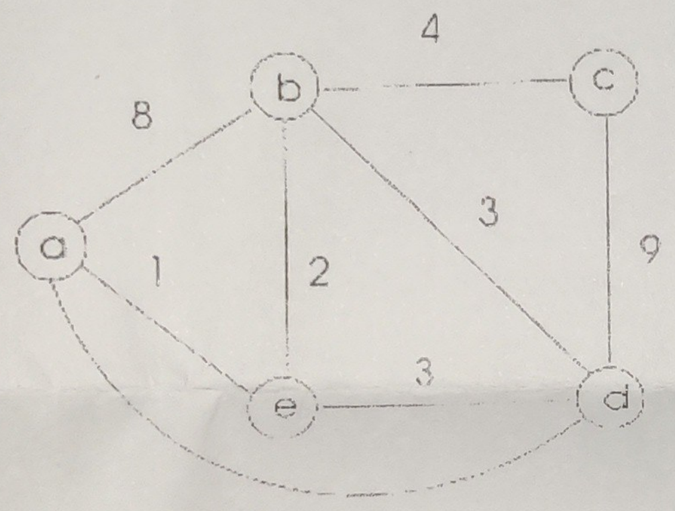
Subject Code:BCS 205

Subject: Design and analysis of algorithms

Time : 1 ½Hours

Maximum Marks : 30

Note: Q. 1 is compulsory.

Q1		(2.5*4)	CO
(a)	Analyse the complexity of the following algorithm function increment(x) if x mod 2==1then return 2 × increment(⌊ x/2 ⌋) else return x + 1		CO1
(b)	Solve the following recurrence using Recursion tree method T(n) = 2T(n/4)+ sqrt(n) , if n>=2 = 1 , Else		CO1
(c)	What are greedy algorithms? What is its limitation.		CO2
(d)	Find the minimum spanning tree of the following graph using Kruskal's Minimum spanning tree algorithm  		CO2

Q2	(Attempt any Two Parts) UNIT-1	(5,5)	CO
(a)	Explain the Strassen's Matrix multiplication algorithm. Analyse its complexity		CO1
(b)	Consider an n – node complete binary tree T . Each node v of T is labeled with a real number Xv. You may assume that the real numbers labeling the nodes are all distinct. A node v of T is a local minimum if the label Xv is less than the label Xv for all nodes w that are joined to v by an edge. Show how to find a local minimum of T using only O(log n) probes to the nodes of T.		CO1
(c)	Give the best case and worst case conditions for quick sort. Write the recurrences correspond to best case and worst case of quick sort. Analyse the worst case, best case and average case complexities of quick sort.		CO1

Q3	(Attempt any Two Parts) UNIT-2	(5,5)	CO
(a)	Explain the greedy algorithm for solving Knapsack problem. Analyse its complexity		CO2

(b)	You are given a sequence of n songs, where the i-th song is li minutes long. You want to place all of the songs onto a collection of CDs, each of which can hold m minutes. Furthermore, (i) The songs must be recorded in the order given: song 1, song 2, . . . , song n. (ii) All songs must be included. (iii) No song may be split across CDs. Give a greedy algorithm to place songs on CDs so as to minimize the number of CDs needed.	CO2
(c)	Explain the greedy algorithm for generating Huffman code and analyse its complexity	CO2

song 1 2 3 4
length l_1 l_2 l_3 l_4

CDs → lm